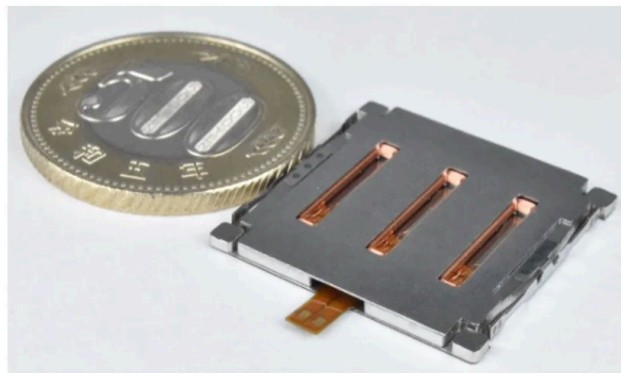


The World's Thinnest Linear Resonant Actuator, TapSense

TapSense[®] designed by NIDEC PRECISION in Tokyo

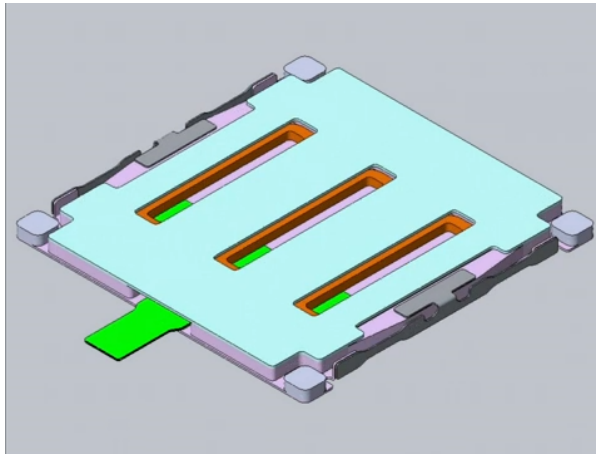
TapSense is Nidec Precision Corporation's permanent magnet-based, world's thinnest* linear resonant actuator. Developed based on the company's lifelong, camera industry-based "precision" manufacturing technology, this **1.4mm-thick** actuator helps make tablets, notebook PCs and other digital devices thinner than their existing models.

*This information is as of April 2024, based on the result of Nidec Precision Corporation's research.



Tap to Experience Realistic Mechanical Feedbacks

TapSense's newly developed magnetic circuit design and spring & damper combination structure realizes both thin form factor and enough vibration force. Furthermore, its high responsiveness makes TapSense easier to control than conventional linear resonant actuator. The high-quality lateral (horizontal) force that this 1.4mm, ultra-thin actuator generates creates a variety of haptic feedbacks, such as tapping, swiping, scrolling, or flicking.



Use TapSense for Multiple Applications

With vibration force of 6.4Gp-p, and with vibration duration of 12ms, TapSense realizes crisp haptic feeling. And with thickness of world-beating* 1.4mm, TapSense helps make tablets, notebook PCs, and other digital devices thinner than their existing models. Further, its wide operating temperature range (-40° C to +85° C) makes TapSense a highly reliable component for automotive panels, wearable medical devices, and so many other mechanical products.



PC/Tablet



Smart Phone



HMI



Smart Watch



VR/Game



Automotive

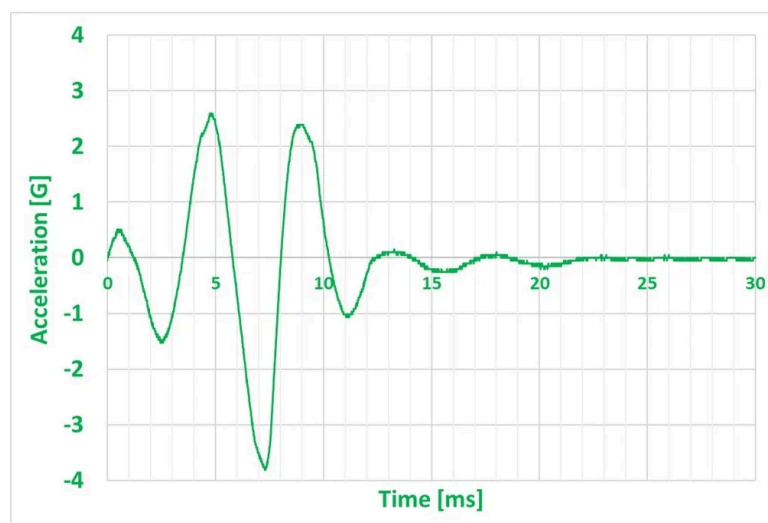
Specification Data

Development Specifications

Size	25 × 25 × 1.4mm
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Resonance frequency	200 Hz
OD* voltage	7.0 Vrms
OD waveform	3-pulse sine wave
OD acceleration @40g jig	6.4 Gp-p
Rise time (0 – 90%)	$\cong$ 30 ms
Fall time (100 – 10%)	$\cong$ 20 ms
Rated voltage	3.0 Vrms
Rated current	360 mA
Resistance value	8 Ω
Operating temperature range	-40 °C ~ +85 °C
*Over Drive	

Example of Vibration Waveform



Contact Information for Inquiries

Sales Department 2, Nidec Precision Corporation

Inquiry by email

NPCJ-TAPS1@nidec.com

- Samples
- Estimation request
- Other inquiries

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